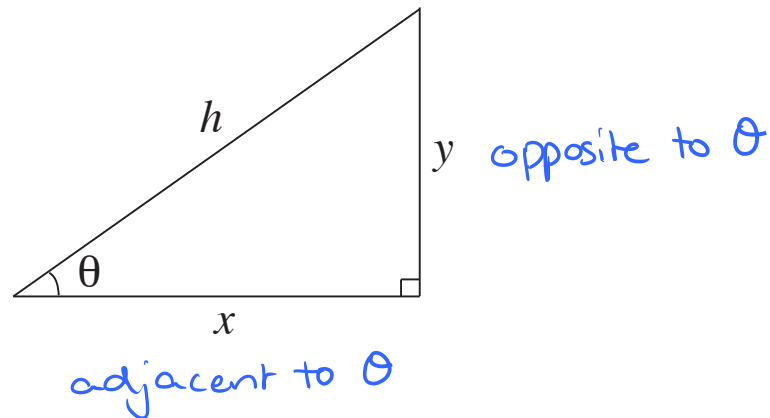


3.5: Trigonometric Functions

Reference Evans 6.1

Consider a right-angled triangle with angle θ and side lengths x , y and h as shown:



The trigonometric functions *sine*, *cosine* and *tangent* of θ are defined as:

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{y}{h}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{x}{h}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{y}{x} = \frac{\sin \theta}{\cos \theta}$$

SOH CAH TOA

Reciprocal Trigonometric Functions

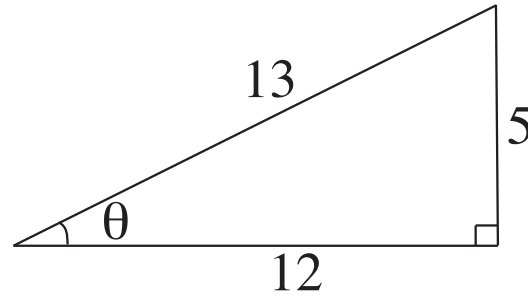
The reciprocal trigonometric functions *secant*, *cosecant* and *cotangent* are defined as:

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cot \theta = \frac{1}{\tan \theta} \text{ or } \frac{\cos \theta}{\sin \theta}$$

Example



For the above right-angled triangle, find:

1. $\sin \theta = 5/13$
2. $\cos \theta = 12/13$
3. $\tan \theta = 5/12$
4. $\sec \theta = 13/12$
5. $\operatorname{cosec} \theta = 13/5$
6. $\cot \theta = 12/5$

You can now attempt Sheet 3 Q19

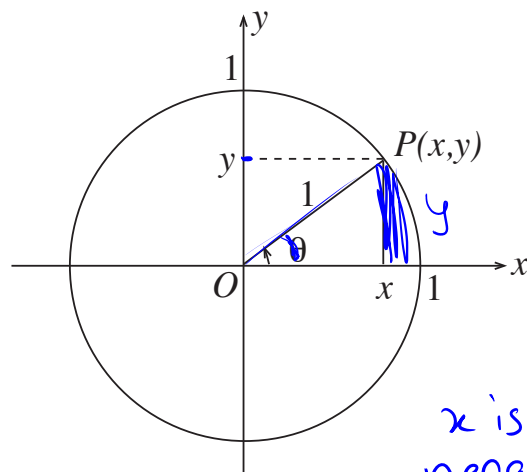
The Unit Circle

We can form right-angled triangles in a *unit circle* (circle of radius 1).

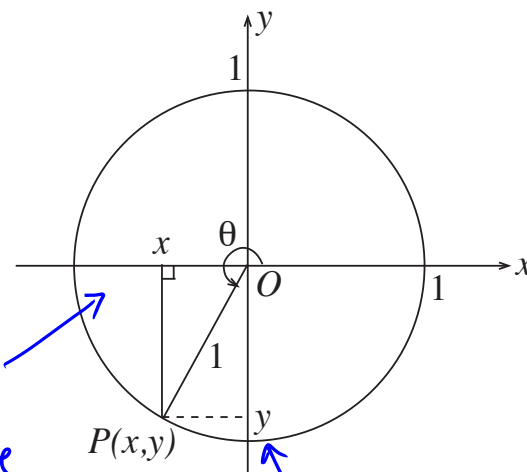
If θ is the anticlockwise angle between the positive x -axis and the ray $\overrightarrow{OP}$ then for all $\theta \in \mathbb{R}$:

$$\sin \theta = \frac{y}{1} = y, \quad \cos \theta = \frac{x}{1} = x \quad \text{and} \quad \tan \theta = \frac{y}{x}$$

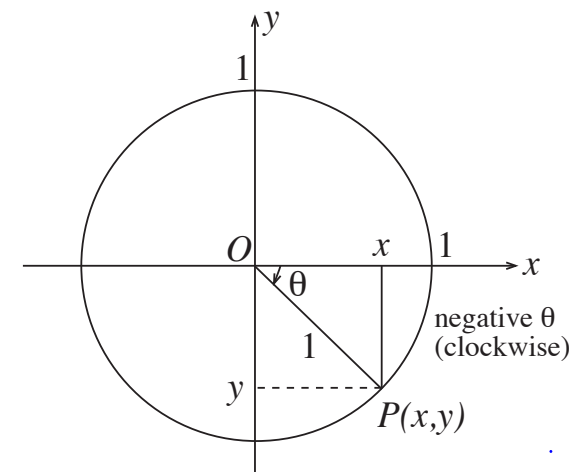
$$\therefore y = \sin \theta \quad x = \cos \theta \quad y/x = \tan \theta.$$



x is negative

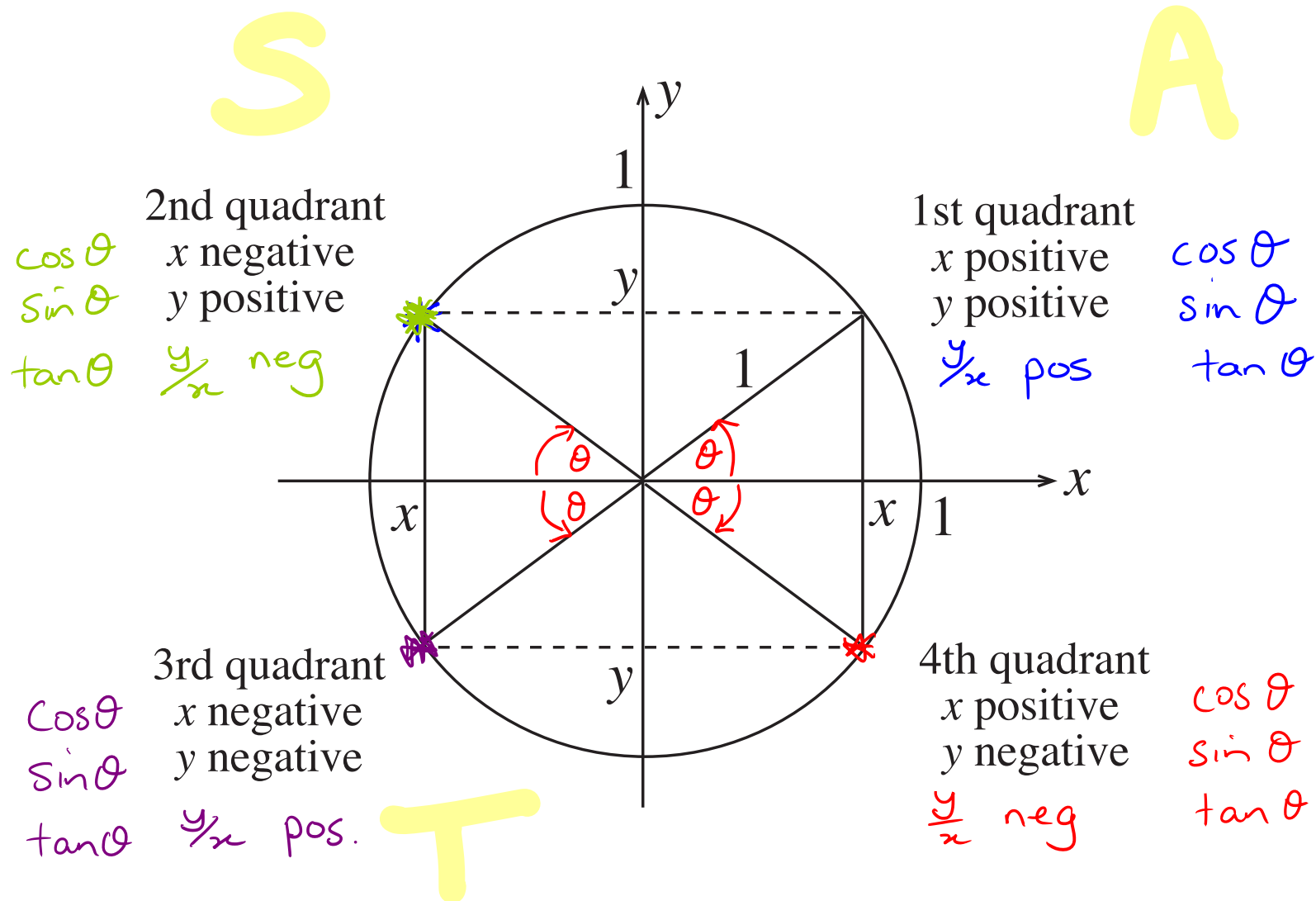


y is negative



Signs of Trigonometric Functions

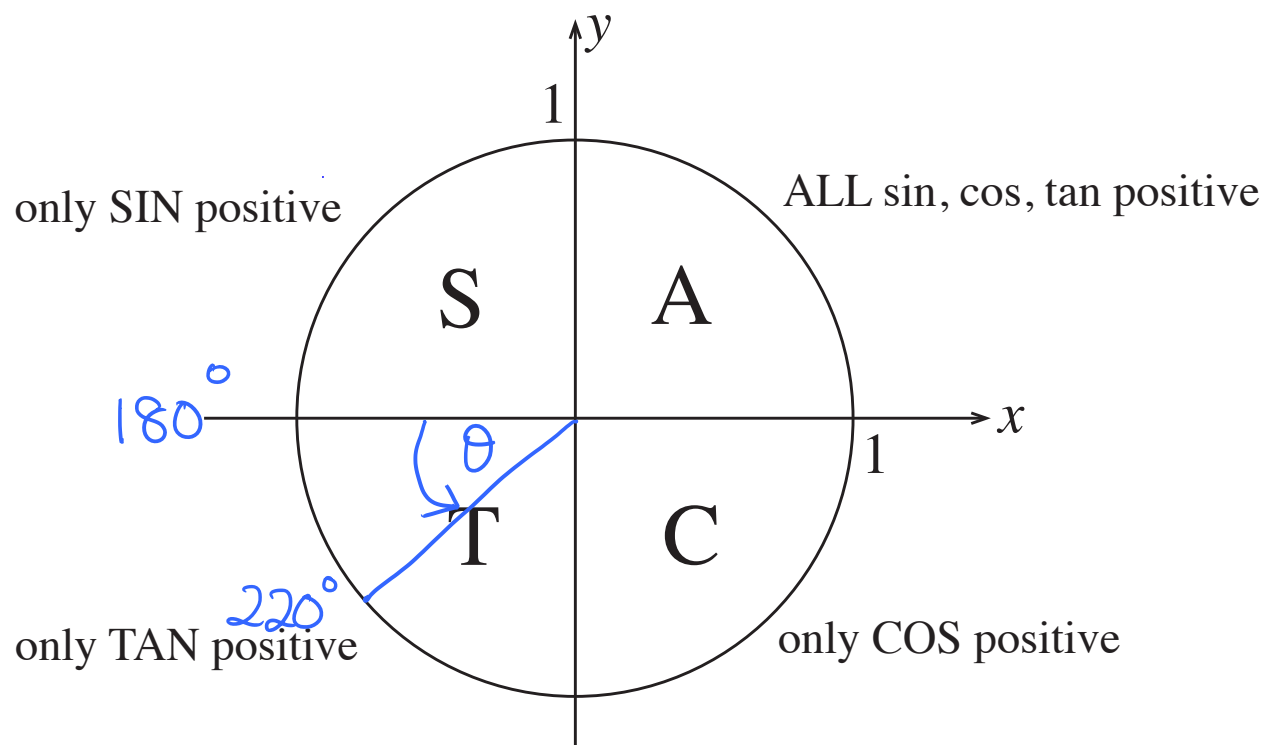
The signs of the trigonometric functions for different values of θ can be determined by noting the signs of x and y in the unit circle:



'CAST' Diagram

Since in the unit circle $\sin \theta = y$, $\cos \theta = x$ and $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{y}{x}$, we obtain the following 'CAST' diagram as a shorthand for remembering signs of trigonometric functions in different parts of the circle:

$$\therefore \cos 220^\circ = -\cos 40^\circ$$



Degrees and Radians

In the unit circle, if we measure the length of the arc from A to B in an anticlockwise direction, we have another way of measuring angles.

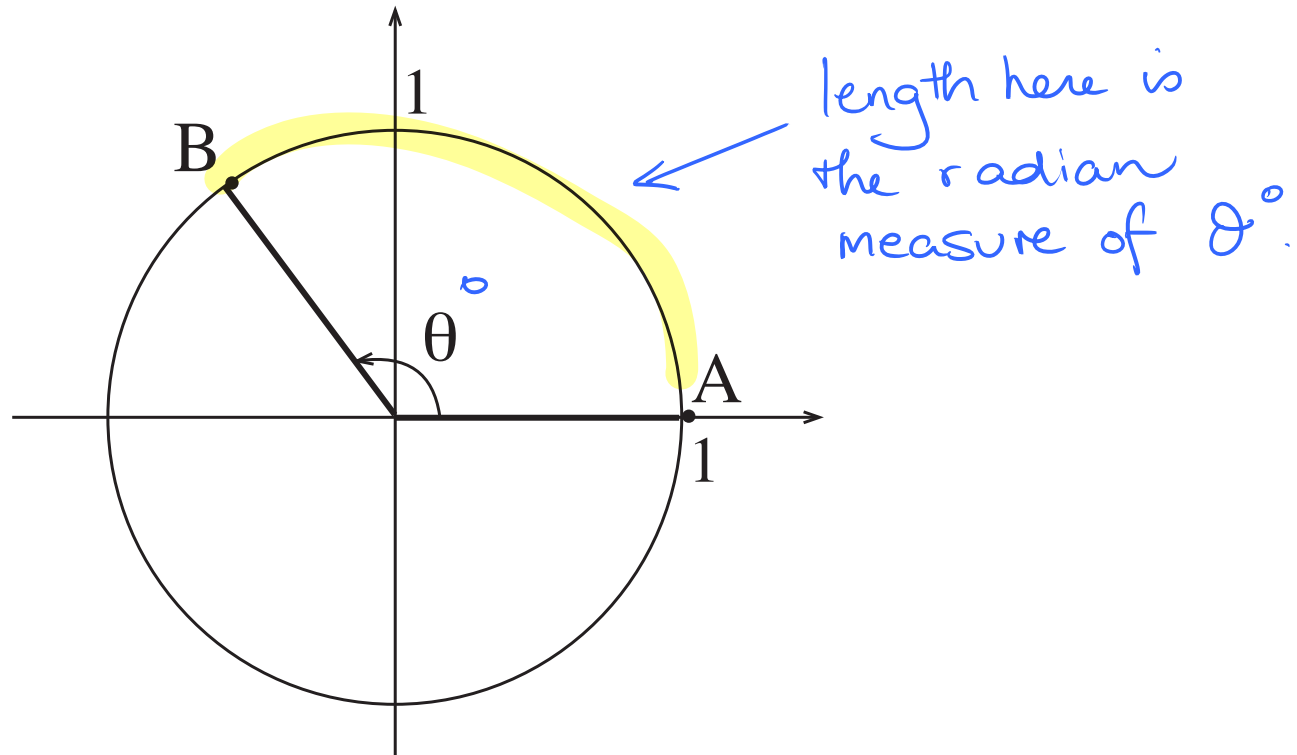
Circumference of
a circle $= 2\pi r$

So for a unit circle

Circumference $= 2\pi$

$$360^\circ \equiv 2\pi$$

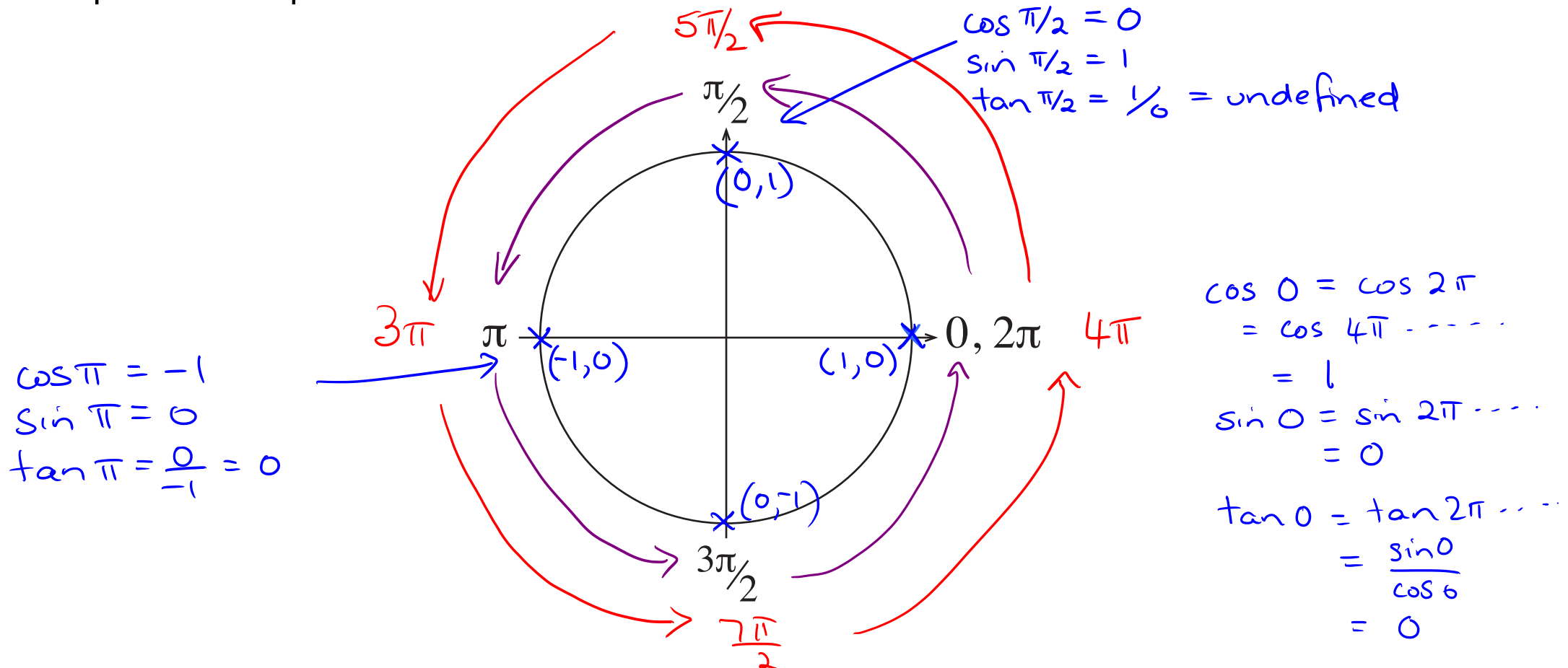
$$180^\circ \equiv \pi$$



This length is called the **radian** measure of the angle θ and we can signify it by the symbol θ^c rather than θ° , but usually we don't show units for radians.

Radians on the Unit Circle

Since the circumference of the unit circle is $2\pi r = 2\pi \cdot 1 = 2\pi$, a complete angle around the whole circle is 2π radians. Hence if we divide the circle into four quarters or quadrants we have radian measures as shown:



Converting Between Degrees and Radians

To convert from degrees to radians multiply by $\frac{\pi}{180}^\circ$

To convert from radians to degrees multiply by $\frac{180^\circ}{\pi}$.

Examples

Convert $360^\circ, 1^\circ, 90^\circ, 135^\circ$ to radians.

$$360^\circ = 2\pi, \quad 1^\circ = 1^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{180}, \quad 90^\circ = 90^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{2}$$
$$135^\circ = 135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}$$

Convert $\pi, 1, \frac{\pi}{3}, \frac{7\pi}{6}$ radians to degrees.

$$\pi = 180^\circ, \quad 1 \times \frac{180}{\pi} = \frac{180}{\pi}^\circ, \quad \frac{\pi}{3} = \frac{\pi}{3} \times \frac{180}{\pi} = 60^\circ$$
$$\frac{7\pi}{6} = \frac{7\pi}{6} \times \frac{180}{\pi} = 210^\circ$$

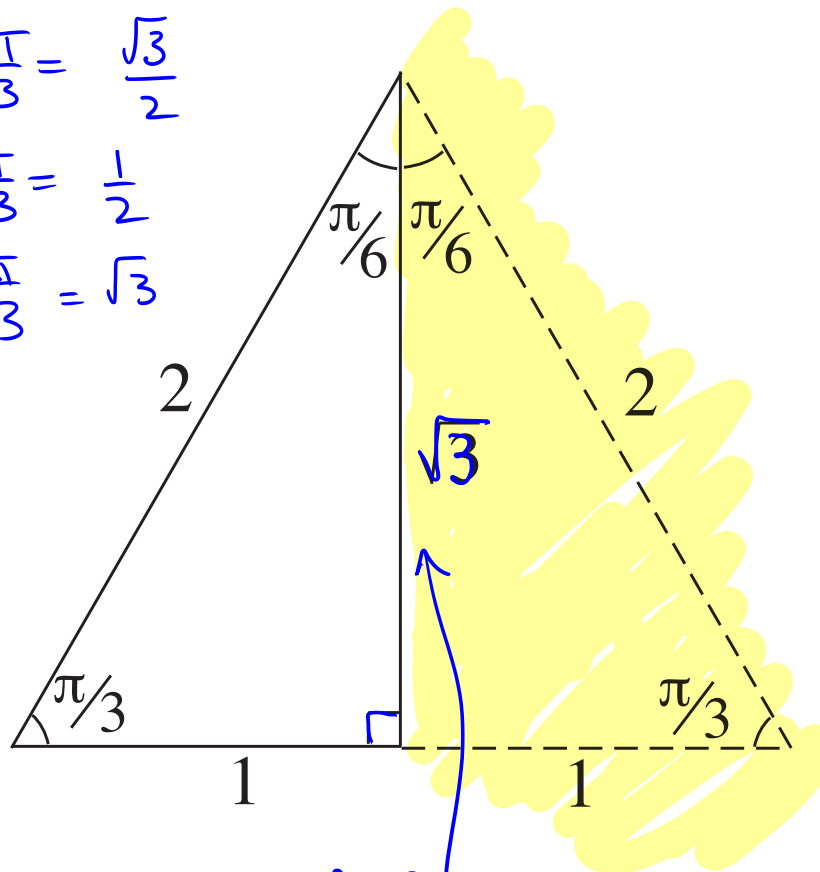
Two Important Triangles

The following triangles help us work out the exact values of **sin**, **cos** and **tan** of certain special angles.

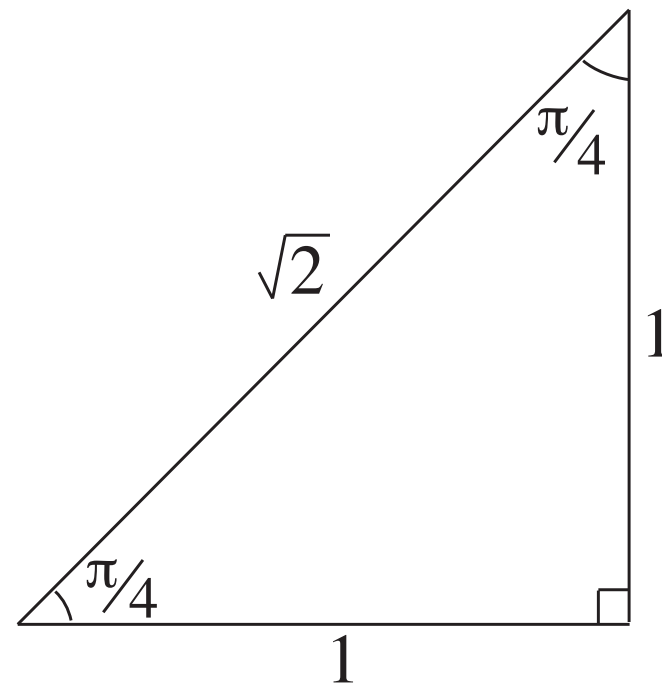
$$\therefore \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

$$\tan \frac{\pi}{3} = \sqrt{3}$$



$$a^2 = 2^2 - 1^2 = 3$$
$$a = \sqrt{3}$$



Exact Values using Triangles

Using the *two important triangles* on the previous slide and your knowledge of the unit circle complete the table with exact values:

purple - from
triangles

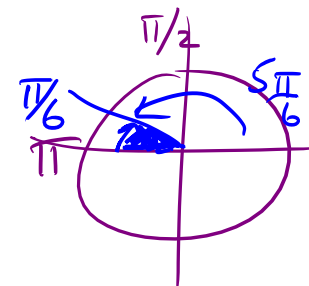
green - from
unit circle

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	undef.
π	0	-1	0
$\frac{3\pi}{2}$	-1	0	undef

***Important to know these ratios, either the triangles or the table

Examples:

Find the exact values of the following trig ratios:



$$1. \cos\left(\frac{5\pi}{6}\right) \overset{Q2}{=} -\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}$$

$$2. \sin\left(\frac{4\pi}{3}\right) \overset{Q3}{=} -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}$$

$$3. \tan\left(\frac{3\pi}{4}\right) \overset{Q2}{=} -\tan\frac{\pi}{4} = -1$$

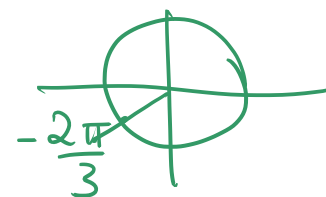
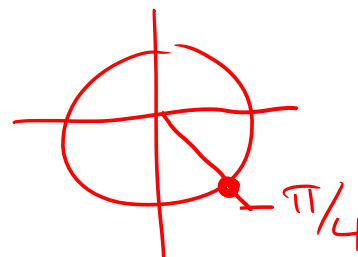
$$4. \sin\left(\frac{11\pi}{6}\right) \overset{Q4}{=} -\sin\frac{\pi}{6} = -\frac{1}{2}$$

$$5. \tan\left(\frac{-\pi}{4}\right) \overset{Q4}{=} -\tan\frac{\pi}{4} = -1$$

$$6. \cos\left(\frac{-2\pi}{3}\right) \overset{Q3}{=} -\cos\frac{\pi}{3} = -\frac{1}{2}$$

Ask yourself:

- quadrant?
- angle from x -axis?
- $\pm$ CAST?
- value of ratio?



Finding Angles

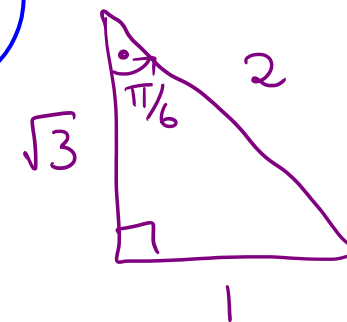
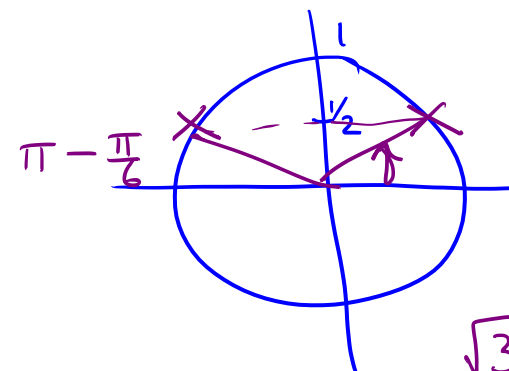
Here we need to work backwards considering both the value of the ratio and its sign, as this determines the quadrants of the resultant angles.

Find all values of θ between 0 and 2π satisfying:

Answers in Q1 + Q2

1. $\sin(\theta) = \frac{1}{2}$

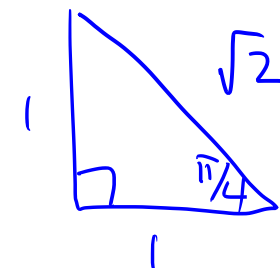
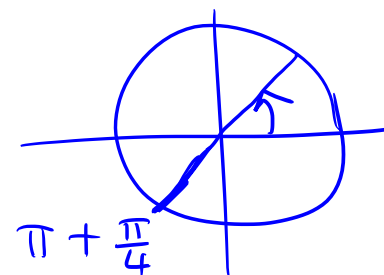
$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$



2. $\tan(\theta) = 1$

Answers in Q1 + Q3

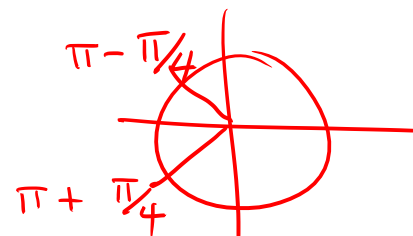
$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$



3. $\cos(\theta) = -\frac{1}{\sqrt{2}}$

Answers in Q2 + Q3

$$\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$$



Solving Trigonometric Equations

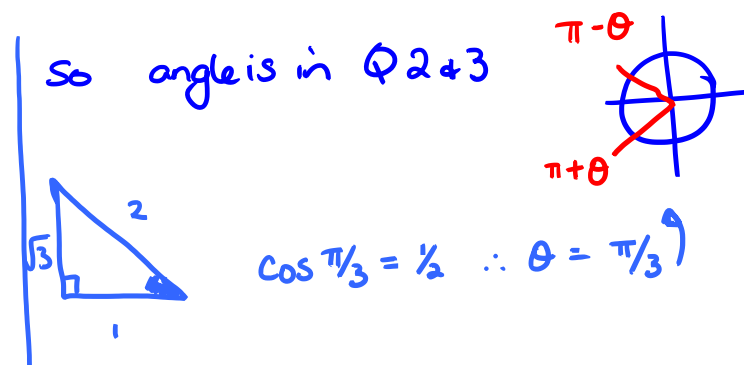
Solve the following trigonometric equations for $x \in [0, 2\pi)$.

1. $\cos x + \frac{1}{2} = 0$

$$\Rightarrow \cos x = -\frac{1}{2}$$

$$\Rightarrow x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$



2. $2\sin^2 x + \sin x - 1 = 0$

Let $a = \sin x$ which means equation becomes

$$2a^2 + a - 1 = 0$$

which can be factorised as

$$(2a - 1)(a + 1) = 0$$

$$\Rightarrow a = \frac{1}{2}, a = -1$$

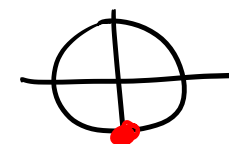
$\therefore \sin x = \frac{1}{2}$ so x is in quadrants 1 and 2 ,

$$x = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\sin x = -1$$

$$x = \frac{3\pi}{2}$$



y value is -1
 $\therefore \sin$ value is -1

$$3. \cos x + \sin x = 0$$

$$\Rightarrow \sin x = -\cos x$$

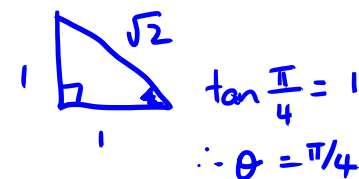
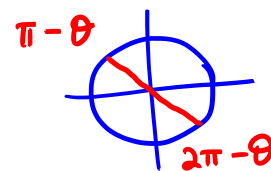
$$\Rightarrow \tan x = -1$$

$$x = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

and dividing by $\cos x$ gives

$\tan$ is negative in Q2 and 4



$$4. 2\tan^2 x - 6 = 0$$

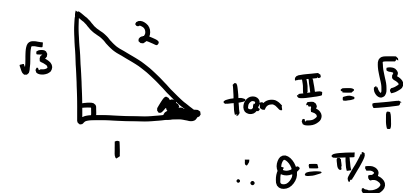
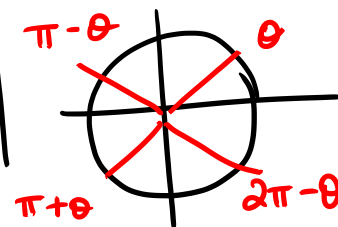
$$\Rightarrow \tan^2 x = 3$$

$$\tan x = \pm \sqrt{3}$$

$$\therefore x = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$\therefore$ we have answers in all four quads.

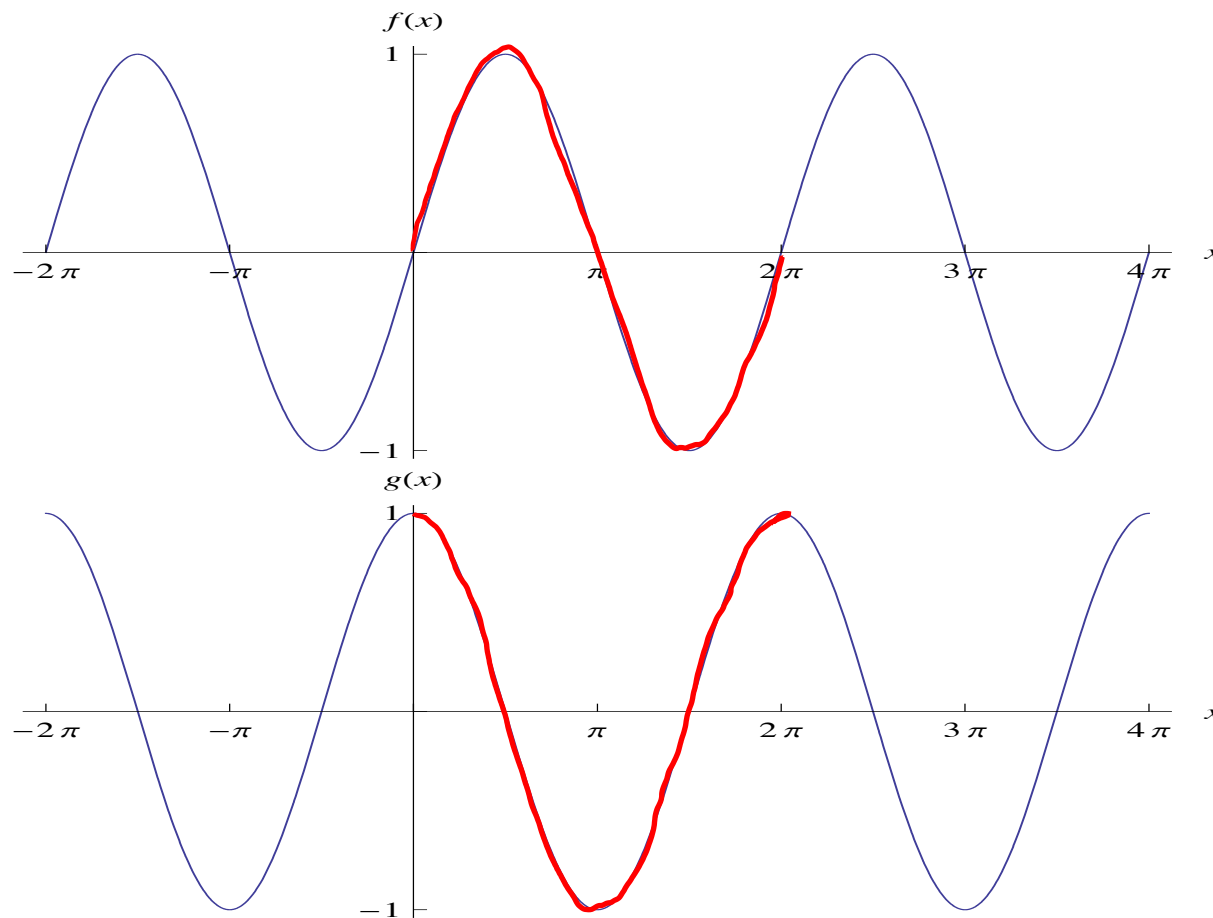


You can now attempt Sheet 3 Q20-21

Sine and Cosine Graphs

Reference Evans 6.2

Plotting the values of $f(x) = \sin x$ and $g(x) = \cos x$ for $x \in \mathbb{R}$ gives the following graphs



The red section is one cycle and note it has a period of 2π and an amplitude of 1.

Properties of Sine and Cosine Graphs

- The sine and cosine functions repeat themselves after an interval (or period) of 2π units. That is:

$$\dots \sin(x - 2\pi) = \sin x = \sin(x + 2\pi) = \sin(x + 4\pi) \dots, \quad \text{and}$$

$$\dots \cos(x - 4\pi) = \cos(x - 2\pi) = \cos x = \cos(x + 2\pi) \dots, \quad \text{for all } x \in \mathbb{R}$$

These functions are therefore said to be *periodic or cyclic*, with *period* 2π .

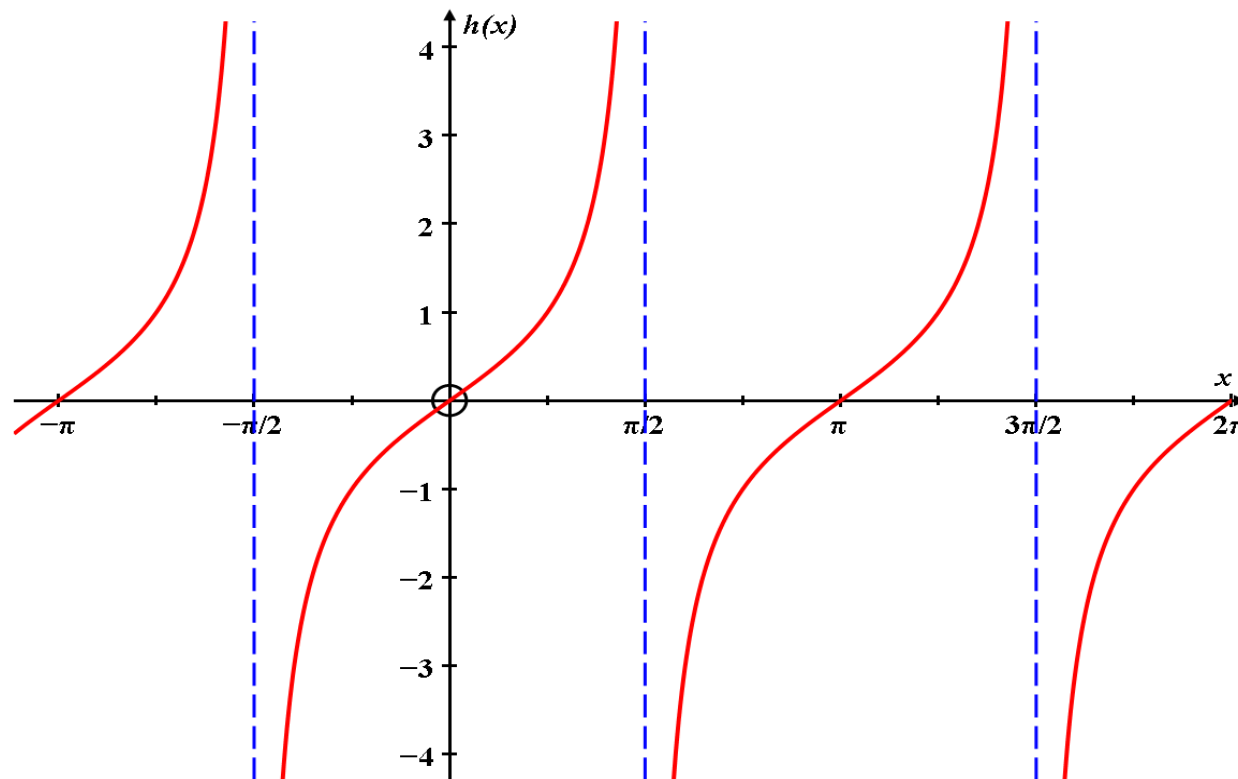
- The maximum and minimum values of $\sin x$ and $\cos x$ are 1 and -1 respectively. The graphs of $f(x) = \sin x$ and $g(x) = \cos x$ are therefore said to have an *amplitude* of 1 .

Graph of Tangent Function

Reference Evans 6.6

The function $h(x) = \tan x = \frac{\sin x}{\cos x}$ is defined for $x \in \mathbb{R}$ such that $\cos x \neq 0$.

Plotting $h(x) = \tan x$ over its domain $x \in \mathbb{R} \setminus \{(2k + 1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$ gives the following graph



So when $\cos x = 0$
we have the
asymptotes for
the $\tan x$
graph.

Properties of Tan Graph

- The tangent function $h(x) = \tan x$ is undefined at $x = \{(2k + 1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$ (this is where $\cos x = 0$).

The graph has vertical *asymptotes* at these x -values, which are usually indicated by dotted or dashed vertical lines.

- The tangent function repeats itself after an interval of π units. That is:

$$\tan \theta = \tan(\theta + \pi), \quad \text{for all } \theta \in \mathbb{R} \setminus \{(2k + 1)\frac{\pi}{2} \mid k \in \mathbb{Z}\}$$

This function therefore has *period* π .

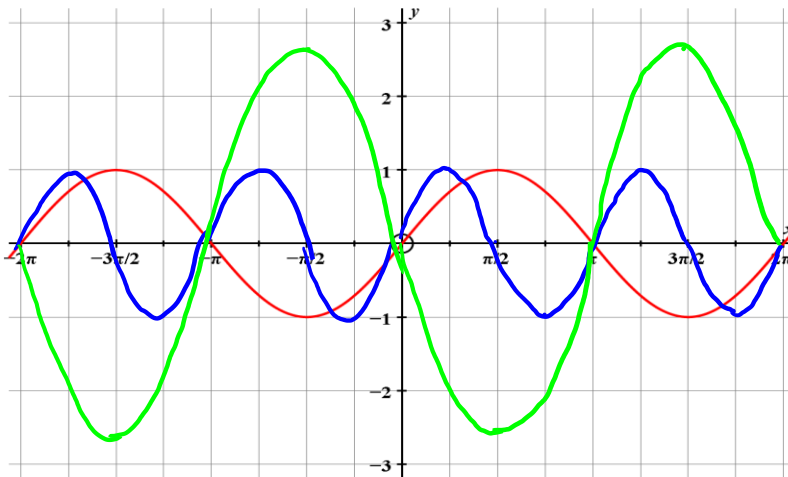
Dilations of Trig Graphs

Reference Evans 6.2, 6.3 & 6.6

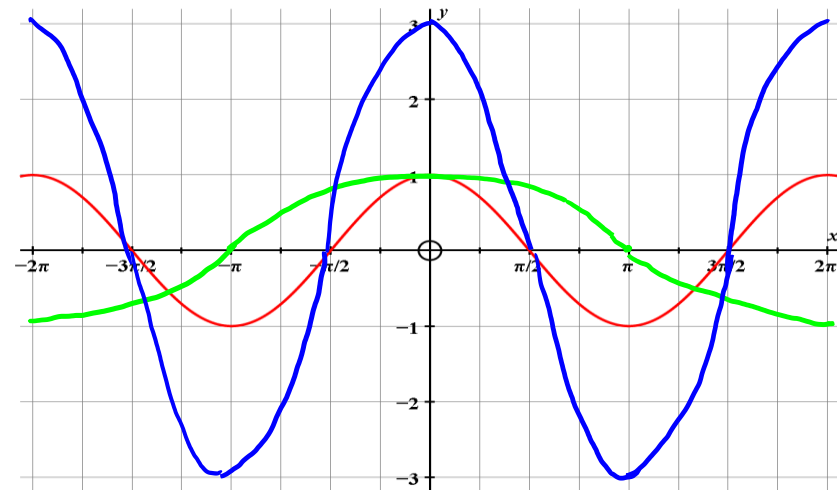
in RED

Sketch the following graphs - the standard sin or cos graph is shown - state the period and amplitude of the transformed graph in each case.

Note: a dilation in the x direction (e.g. $y = \cos bx$) changes the period (to $\frac{2\pi}{b}$) and one in the y direction (e.g. $y = a \sin x$) changes the amplitude (to a)



1. $y = \sin 2x$ So period = $\frac{2\pi}{2} = \pi$, amp = 1
3. $y = -\frac{5}{2} \sin x$ period = 2π , amp = $\frac{5}{2} = 2.5$
- reflected in x axis.



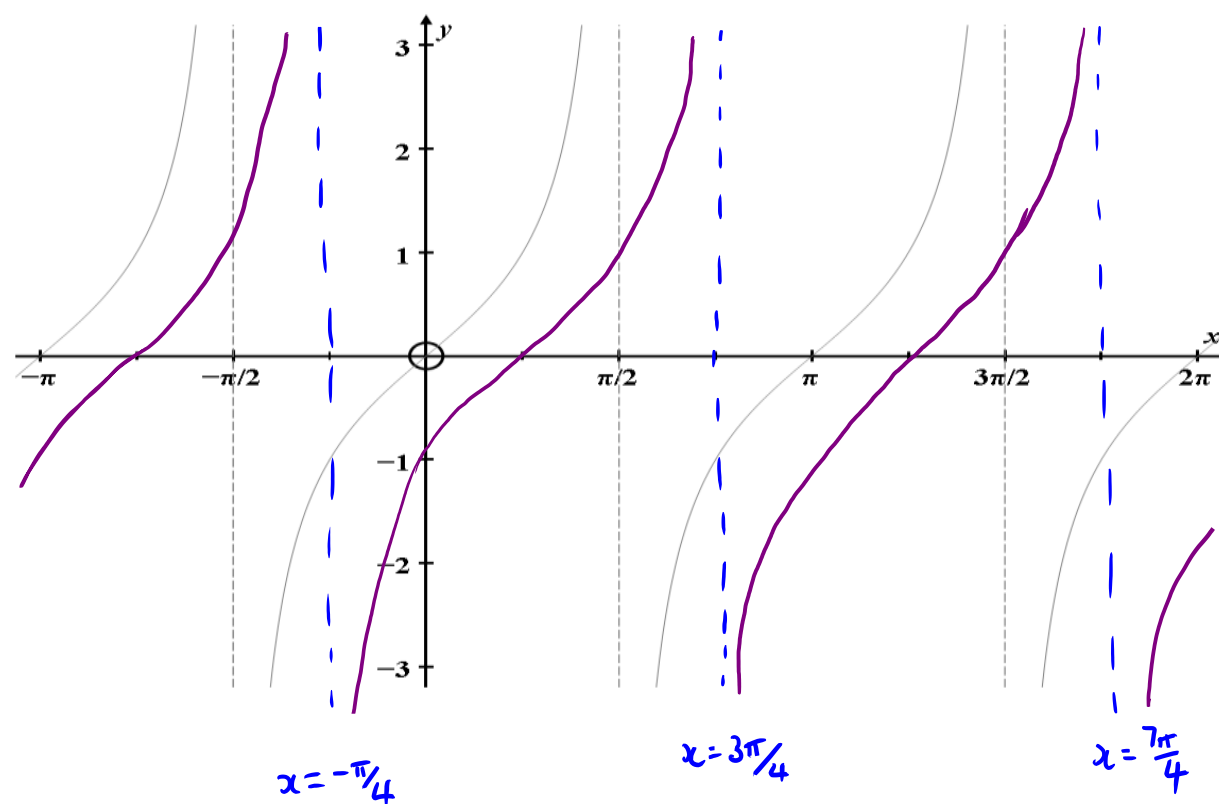
2. $y = 3 \cos x$ period = 2π , amp = 3
4. $y = \cos\left(\frac{x}{2}\right)$ period = $\frac{2\pi}{\frac{1}{2}} = 4\pi$, amp = 1

Translations of Trigonometric Graphs

Sketch the following graphs on the axes below –adding or subtracting a value to the trig function will translate the graph up or down, while adding or subtracting a value to the x term will translate the graph to the left or right respectively.

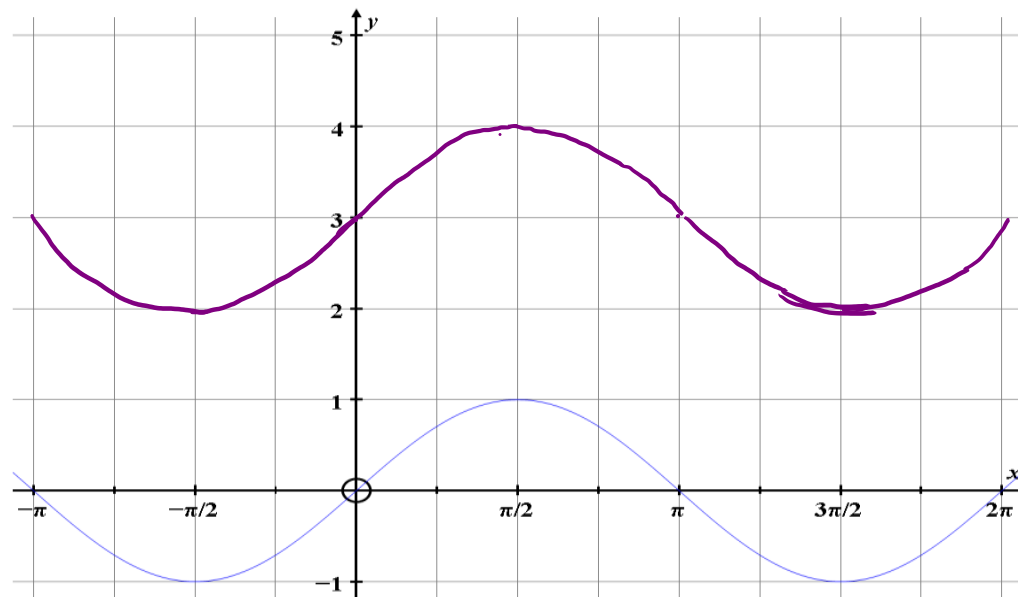
1. $y = \tan(x - \frac{\pi}{4})$

This will translate the
tan graph $\frac{\pi}{4}$ units to
the right.



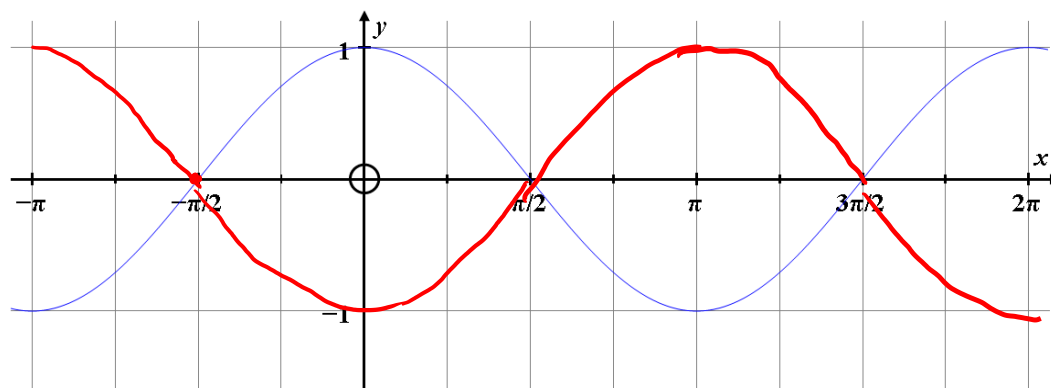
2. $y = \sin(x) + 3$

This translates the $\sin x$ graph $\uparrow$ 3 units



3. $y = \cos(x + \pi)$

This translates the $\cos x$ graph π units to the left.



Multiple Transformations

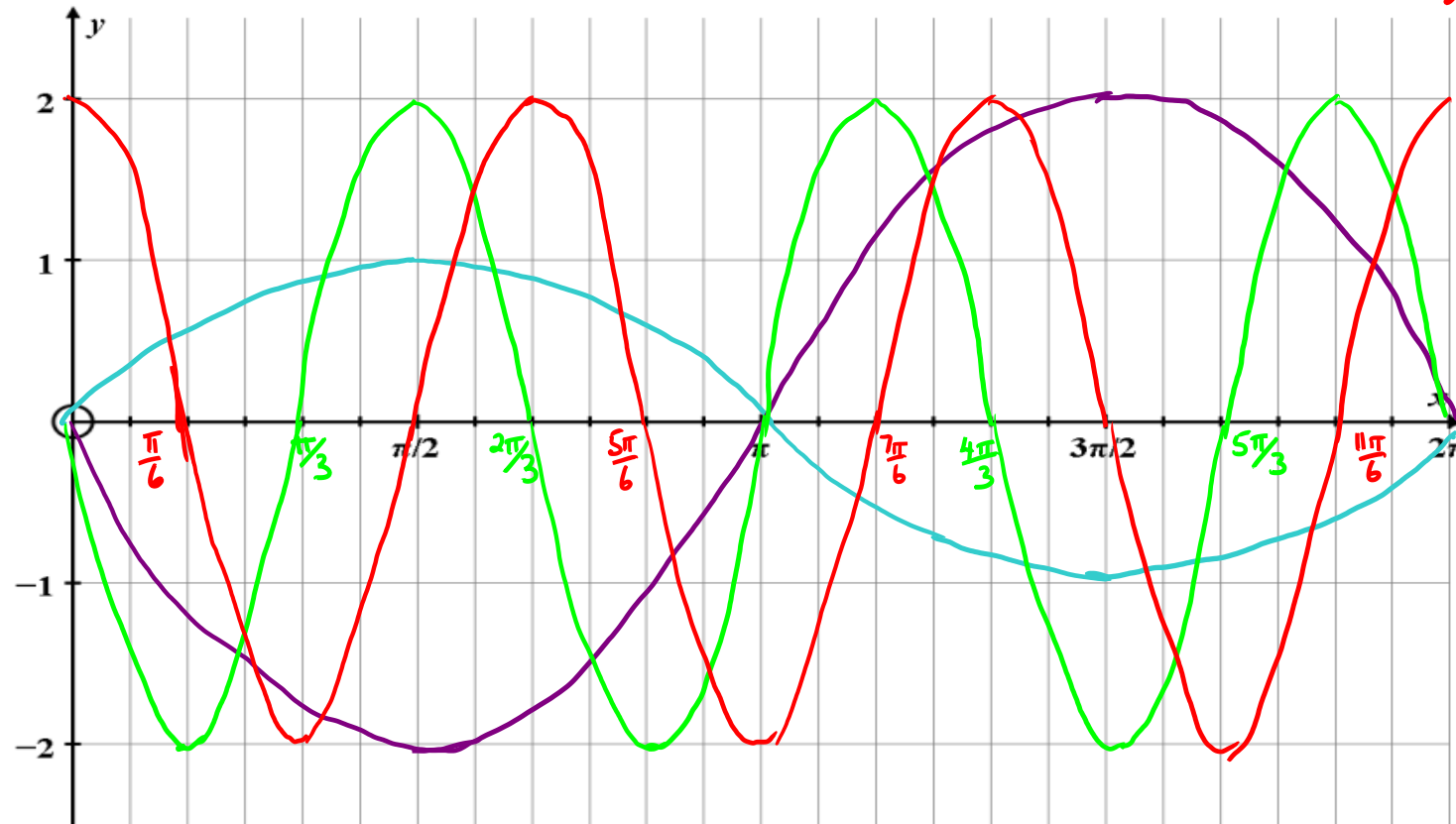
Sketch the following sequence of graphs on the same set of axes.

$y_1 = \sin x$

$y_2 = -2 \sin x$

$y_3 = -2 \sin 3x$

$y_4 = -2 \sin \left(3x - \frac{\pi}{2} \right)$
 $= -2 \sin 3 \left(x - \frac{\pi}{6} \right)$



y_3 - period = $\frac{2\pi}{3}$
 y_2
 y_1
 y_4 - y_3 graph translated $\frac{\pi}{6}$ to right.

You can now attempt Sheet 3 Q22-23

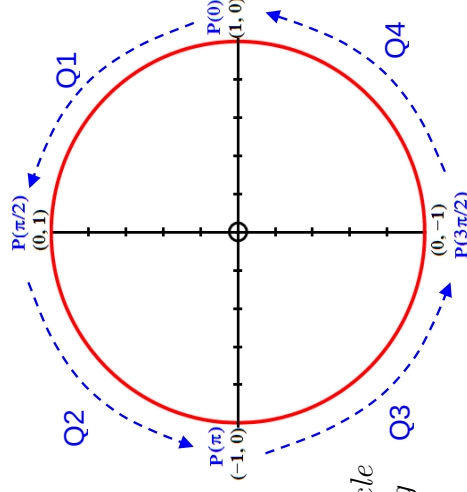
REVISION - TRIGONOMETRY

A: Finding trig ratios in the Unit Circle

1. Identify the quadrant that the angle is in:

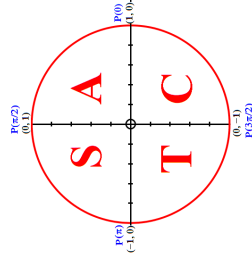
- Q1 has angles from $0 \rightarrow \frac{\pi}{2}$
- Q2 has angles from $\frac{\pi}{2} \rightarrow \pi$
- Q3 has angles from $\pi \rightarrow \frac{3\pi}{2}$
- Q4 has angles from $\frac{3\pi}{2} \rightarrow 2\pi$

*This is just the first revolution of the unit circle
We can of course find bigger angles by moving
around the circle more than once or negative
angles by going in the opposite direction*



2. Decide if the ratio you need to find (usually sin, cos or tan) is positive or negative in the quadrant you found in step 1

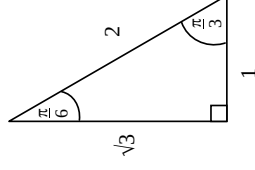
(most students use CAST to remember the signs)



3. Use the special triangles to find the ratio required

(remember SOH-CAH-TOA)

e.g. $\sin \frac{\pi}{6} = \frac{1}{2}, \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$



4. For angles that give points on the x or y axes we use the basic definitions:

- $\cos \theta$ is the **x coordinate**
- $\sin \theta$ is the **y coordinate**
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Therefore we can find values like:

$$\cos \pi = -1 \quad \text{or} \quad \sin \left(-\frac{\pi}{2}\right) = -1 \quad \text{or} \quad \tan \frac{3\pi}{2} = \frac{\left(\sin \frac{3\pi}{2}\right)}{\left(\cos \frac{3\pi}{2}\right)} = \frac{-1}{0} = \text{undefined}$$

It may help to look at the unit circle above to see these points (and remember that $-\frac{\pi}{2}$ and $\frac{3\pi}{2}$ are just different names for the same point on the circle.

PRACTICE EXERCISE A

1. Find the values of the following trig ratios - the steps mentioned on the previous page have been spelt out for the first few questions (then you are on your own to do the rest by following the same process):

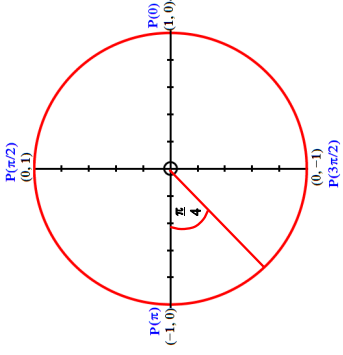
(a) $\sin \frac{5\pi}{4}$

i. $\frac{5\pi}{4}$ is in $Q \dots\dots$ (as $\frac{5\pi}{4} = \pi + \frac{\pi}{4}$)

ii. In $Q \dots\dots$ sin is $\dots\dots$ (+ or -)

iii. From triangle 2 we know that $\sin \frac{\pi}{4} = \dots\dots$

$\implies \sin \frac{5\pi}{4} =$



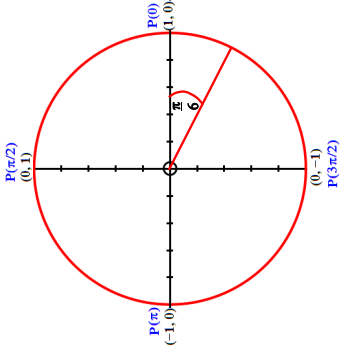
(b) $\cos \frac{11\pi}{6}$

i. $\frac{11\pi}{6}$ is in $Q \dots\dots$ (as $\frac{11\pi}{6} = 2\pi - \frac{\pi}{6}$)

ii. In $Q \dots\dots$ cos is $\dots\dots$ (+ or -)

iii. From triangle 1 we know that $\cos \frac{\pi}{6} = \dots\dots$

$\implies \cos \frac{11\pi}{6} =$



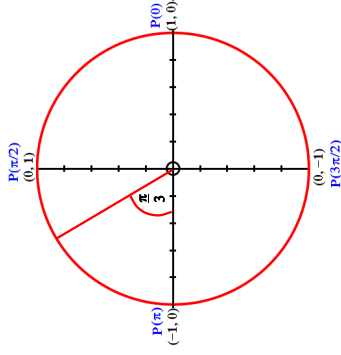
(c) $\tan \frac{8\pi}{3}$

i. $\frac{8\pi}{3}$ is in $Q \dots\dots$ (as $\frac{8\pi}{3} = 3\pi - \frac{\pi}{3} = \pi - \frac{\pi}{3}$)

ii. In $Q \dots\dots$ tan is $\dots\dots$ (+ or -)

iii. From triangle 1 we know that $\tan \frac{\pi}{3} = \dots\dots$

$\implies \tan \frac{8\pi}{3} =$



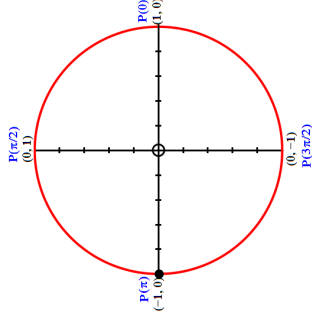
(d) $\cos 5\pi$

i. 5π is on the $\dots\dots$ axis (x or y)

ii. It has coordinates $(\dots, \dots)$

iii. As cos is the x coordinate we know:

$\implies \cos 5\pi =$



2. Now try the same process with these questions:

(a) $\sin(\frac{5\pi}{3})$

(b) $\cos(\frac{5\pi}{4})$

(c) $\tan(\frac{7\pi}{6})$

(d) $\sin(2\pi)$

(e) $\cos(-\frac{7\pi}{2})$

(f) $\sin(-\frac{7\pi}{4})$

(g) $\tan(-\frac{17\pi}{6})$

There are other trig ratios we can use but they are based on the standard ones. So to find cosec, sec or cot we calculate sin, cos or tan respectively and then just "turn them upside down" because of the definitions:

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \cot \theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$$

Examples: We found on page 1 in section 3 that:

$$\sin \frac{\pi}{6} = \frac{1}{2} \implies \operatorname{cosec} \frac{\pi}{6} = \frac{2}{1} = 2 \quad \text{and}$$

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2} \implies \sec \frac{\pi}{6} = \frac{2}{\sqrt{3}} \quad (\text{or } \frac{2\sqrt{3}}{3})$$

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \implies \cot \frac{\pi}{6} = \sqrt{3}$$

3. Find the following trig ratios (you may find your answers to Q2 useful in some cases):

(a) $\operatorname{cosec}(\frac{5\pi}{3})$

(b) $\sec(-\frac{7\pi}{2})$

(c) $\cot(\frac{7\pi}{6})$

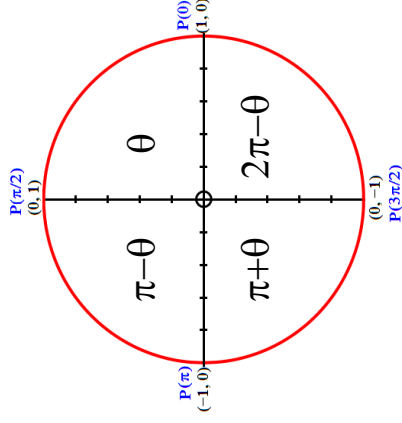
(d) $\sec(\frac{3\pi}{4})$

(e) $\cot(6\pi)$

B: Solving Trig equations

1. Rearrange the equation to make the trig ratio (usually sin, cos or tan) the subject
 e.g. $\sin x + 1 = 0 \implies \sin x = -1$
 $\sqrt{2} \cos(x + \frac{5\pi}{6}) - 1 = 0 \implies \cos(x + \frac{5\pi}{6}) = \frac{1}{\sqrt{2}}$
2. Find the basic angle that satisfies this ratio – *this may involve looking at the angles in the two special triangles or looking at the coordinates of points on the unit circle where they intersect with the two axes.*
3. Decide which quadrants the answers must be in - look at the sign of the trig ratio
 e.g. if the sin ratio has a negative answer then angles must be in Q3 and Q4
 if the cos ratio has a positive answer then angles must be in Q1 and Q4
so we are really looking at our CAST diagram "backwards"

4. Use knowledge of the unit circle to find the basic angle in the right quadrants
 So the unit circle on the right might be useful here
 Note that θ is the basic angle found in step 2
5. Check the domain of the question
have you found all the solutions?



you may need to add or subtract 2π if you need bigger angles or negative angles for the domain given in the question - that is, angles outside the standard $0 \rightarrow 2\pi$ which is only one revolution around the unit circle

PRACTICE EXERCISE B

Solve the following trig equations over the given domains (as in Exercise A you are guided through the first few questions then you should use the same process to complete the rest):

$$1. 2 \cos x = \sqrt{3} \quad x \in (0, 2\pi)$$

$$\cos x = \frac{\dots}{\dots} \quad \left| \begin{array}{l} \text{basic angle } \theta = \frac{\pi}{6} \quad \text{see special triangle 1} \\ \text{cos is + in Q...}(\theta) \text{ and Q...}(2\pi - \theta) \end{array} \right.$$

$$\implies x = \frac{\pi}{6}, \quad 2\pi - \frac{\pi}{6}$$

$$\implies x = \frac{\pi}{6}, \quad \ddots \frac{\pi}{6}$$

$$\left| \text{ (mentalcheck: these values are in the domain } 0 \rightarrow 2\pi) \right.$$

Note: if we added or subtracted 2π to either of our answers we would get values outside the domain, so we don't need to do anything else here

$$2. \tan(x - \frac{\pi}{4}) - \sqrt{3} = 0 \quad x \in (0, 2\pi)$$

$$\tan(x - \frac{\pi}{4}) = \dots$$

$$\implies x - \frac{\pi}{4} = \dots, \dots$$

$$\implies x - \frac{\pi}{4} = \dots, \dots$$

$$\implies x = \dots + \frac{\pi}{4}, \dots + \frac{\pi}{4}$$

$$\implies x = \dots, \dots$$

basic angle $\theta = \dots$ see special triangle 1
 tan is $\dots (+/-)$ in Q...(θ) and Q3 (...)

add $\frac{\pi}{4}$ to RHS of equation to find x
 use LCD to add fractions

(mental check: *these values are inside the domain of $0 \rightarrow 2\pi$, and if we added or subtracted 2π to either of them we would get answers outside the domain, so we don't need to do anything else here)*

$$3. 8 \sin(x + \frac{\pi}{6}) + 4 = 0 \quad x \in (-2\pi, 2\pi)$$

$$\sin(x + \frac{\pi}{6}) = \dots$$

$$\implies x + \frac{\pi}{6} = \dots, \dots$$

$$\implies x + \frac{\pi}{6} = \dots, \dots$$

$$\implies x = \dots - \frac{\pi}{6}, \dots - \frac{\pi}{6}$$

$$\implies x = \dots, \dots$$

$$\text{and } x = \dots - 2\pi, \dots - 2\pi$$

basic angle $\theta = \dots$ see special triangle 1
 sin is $-$ in Q3 ($\pi + \theta$) and Q...(...)

subtract $\frac{\pi}{6}$ to RHS of equation to find x
 use LCD to add fractions
 domain needs negative angles too

Note: we need all four answers – the two negative and the original two positive ones

So the final solutions are:

$$x = \dots, \dots, \dots, \dots$$

$$4. \sqrt{2} \cos x + 1 = 0 \quad x \in (-2\pi, 2\pi)$$

$$5. \sin(x + \frac{\pi}{3}) = 1 \quad x \in (0, 4\pi]$$

$$6. \cos(x - \frac{3\pi}{4}) = 0 \quad x \in (0, 2\pi)$$

$$7. 2 \cos^2 x + 3 \cos x + 1 = 0 \quad x \in [0, 2\pi]$$

ANSWERS

PRACTICE EXERCISE A

- Find the values of the following trig ratios - the steps mentioned on the previous page have been spelt out for the first few questions (then you are on your own to do the rest by following the same process):

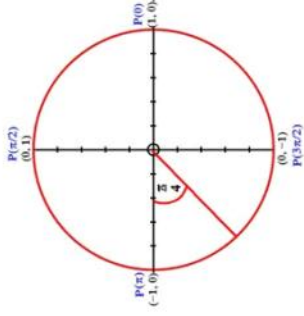
(a) $\sin \frac{5\pi}{4}$

i. $\frac{5\pi}{4}$ is in Q **3** (as $\frac{5\pi}{4} = \pi + \frac{\pi}{4}$)

ii. In Q **3**...sin is **-** (+ or -)

iii. From triangle 2 we know that $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin \frac{5\pi}{4} = \boxed{-\frac{1}{\sqrt{2}}}$



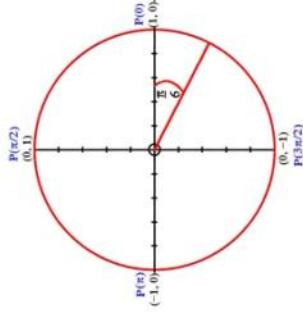
(b) $\cos \frac{11\pi}{6}$

i. $\frac{11\pi}{6}$ is in Q **4** (as $\frac{11\pi}{6} = 2\pi - \frac{\pi}{6}$)

ii. In Q **4**...cos is **+** (+ or -)

iii. From triangle 1 we know that $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$\Rightarrow \cos \frac{11\pi}{6} = \boxed{\frac{\sqrt{3}}{2}}$



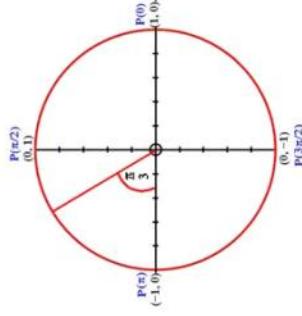
(c) $\tan \frac{8\pi}{3}$

i. $\frac{8\pi}{3}$ is in Q **2** (as $\frac{8\pi}{3} = 3\pi - \frac{\pi}{3} = \pi - \frac{\pi}{3}$)

ii. In Q **2**...tan is **-** (+ or -)

iii. From triangle 1 we know that $\tan \frac{\pi}{3} = \sqrt{3}$

$\Rightarrow \tan \frac{8\pi}{3} = \boxed{-\sqrt{3}}$



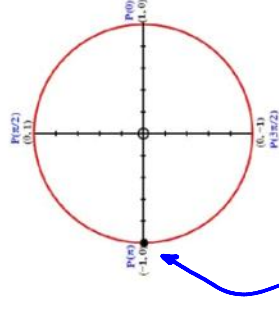
(d) $\cos 5\pi$

i. 5π is on the **x** axis (x or y)

ii. It has coordinates **(-1, 0)**

iii. As cos is the x coordinate we know:

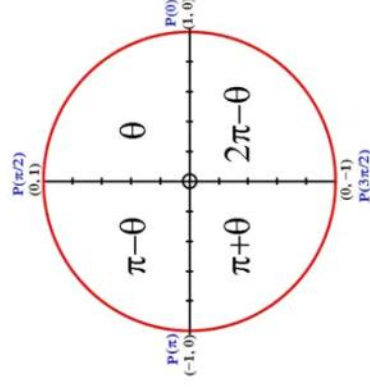
$\Rightarrow \cos 5\pi = \boxed{-1}$



$\pi, 3\pi, 5\pi, 7\pi, \dots$
are all in same position

B: Solving Trig equations

- Rearrange the equation to make the trig ratio (usually $\sin$, $\cos$ or $\tan$) the subject
 e.g. $\sin x + 1 = 0 \implies \sin x = -1$
 $\sqrt{2} \cos(x + \frac{5\pi}{6}) - 1 = 0 \implies \cos(x + \frac{5\pi}{6}) = \frac{1}{\sqrt{2}}$
- Find the basic angle that satisfies this ratio – *this may involve looking at the angles in the two special triangles or looking at the coordinates of points on the unit circle where they intersect with the two axes.*
- Decide which quadrants the answers must be in - look at the sign of the trig ratio
 e.g. if the $\sin$ ratio has a negative answer then angles must be in Q3 and Q4
 if the $\cos$ ratio has a positive answer then angles must be in Q1 and Q4
so we are really looking at our CAST diagram "backwards"
- Use knowledge of the unit circle to find the basic angle in the right quadrants
 So the unit circle on the right might be useful here
 Note that θ is the basic angle found in step 2
- Check the domain of the question
have you found all the solutions?



you may need to add or subtract 2π if you need bigger angles or negative angles for the domain given in the question - that is, angles outside the standard $0 \rightarrow 2\pi$ which is only one revolution around the unit circle

PRACTICE EXERCISE B

Solve the following trig equations over the given domains (as in Exercise A you are guided through the first few questions then you should use the same process to complete the rest):

$$1. 2 \cos x = \sqrt{3} \quad x \in (0, 2\pi)$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$\implies x = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\implies x = \frac{\pi}{6}, \underline{11\pi}{6}$$

basic angle $\theta = \frac{\pi}{6}$ see special triangle 1
 $\cos$ is + in Q. **1**. (θ) and Q. **4**. ($2\pi - \theta$)

(mentalcheck): these values are in the domain $0 \rightarrow 2\pi$)

Note: if we added or subtracted 2π to either of our answers we would get values outside the domain, so we don't need to do anything else here

$$2. \tan(x - \frac{\pi}{4}) - \sqrt{3} = 0 \quad x \in (0, 2\pi)$$

$$\tan(x - \frac{\pi}{4}) = \sqrt{3}$$

$$\Rightarrow x - \frac{\pi}{4} = \frac{\pi}{3}, \quad \pi + \frac{\pi}{3}$$

$$\Rightarrow x - \frac{\pi}{4} = \frac{\pi}{3}, \quad \frac{4\pi}{3}$$

$$\Rightarrow x = \frac{\pi}{3} + \frac{\pi}{4}, \quad \frac{4\pi}{3} + \frac{\pi}{4}$$

$$\Rightarrow x = \frac{7\pi}{12}, \quad \frac{19\pi}{12}$$

(mental check: these values are inside the domain of 0 to 2π, and if we added or subtracted 2π to either of them we would get answers outside the domain, so we don't need to do anything else here)

$$3. 8 \sin(x + \frac{\pi}{6}) + 4 = 0 \quad x \in (-2\pi, 2\pi)$$

$$\sin(x + \frac{\pi}{6}) = -\frac{4}{8} = -\frac{1}{2}$$

$$\Rightarrow x + \frac{\pi}{6} = \frac{\pi}{6}, \quad 2\pi - \frac{\pi}{6}$$

$$\Rightarrow x + \frac{\pi}{6} = \frac{7\pi}{6}, \quad \frac{11\pi}{6}$$

$$\Rightarrow x = \frac{7\pi}{6} - \frac{\pi}{6}, \quad \frac{11\pi}{6} - \frac{\pi}{6}$$

$$\Rightarrow x = \pi, \quad \frac{5\pi}{3}$$

$$\text{and } x = \pi \dots - 2\pi, \quad \frac{5\pi}{3} \dots - 2\pi$$

Note: we need all four answers – the two negative and the original two positive ones

So the final solutions are:

$$x = \dots \pi, \quad -\frac{\pi}{3}, \quad \pi, \quad \frac{5\pi}{3}, \dots$$

$$4. \sqrt{2} \cos x + 1 = 0 \quad x \in (-2\pi, 2\pi)$$

$$5. \sin(x + \frac{\pi}{3}) = 1 \quad x \in (0, 4\pi]$$

$$6. \cos(x - \frac{3\pi}{4}) = 0 \quad x \in (0, 2\pi)$$

$$7. 2 \cos^2 x + 3 \cos x + 1 = 0 \quad x \in [0, 2\pi]$$

See next page

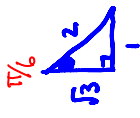


basic angle $\theta = \frac{\pi}{3}$ see special triangle 1
tan is $\frac{1}{1}$ in Q1 (θ) and Q3 ($\pi + \theta$.)

add $\frac{\pi}{4}$ to RHS of equation to find x

use LCD to add fractions

($0 \rightarrow \frac{34\pi}{12}$)



basic angle $\theta = \frac{\pi}{6}$ see special triangle 1
sin is $-\frac{1}{2}$ in Q3 ($\pi + \theta$) and Q4 ($2\pi - \theta$.)

subtract $\frac{\pi}{6}$ to RHS of equation to find x

use LCD to add fractions

domain needs negative angles too

4) $\sqrt{2} \cos x + 1 = 0$ $x \in (-2\pi, 2\pi)$ negative revolution $\therefore -2\pi$

$$\Rightarrow \cos x = -\frac{1}{\sqrt{2}}$$

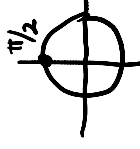
$$\Rightarrow x = \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}$$

$$\rightarrow x = \frac{3\pi}{4}, \frac{5\pi}{4}$$

and $x = \frac{3\pi}{4} - 2\pi, \frac{5\pi}{4} - 2\pi$

So final solutions are:

$$x = -\frac{5\pi}{4}, -\frac{3\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$$



5) $\sin(x + \pi/3) = 1$ $x \in (0, 4\pi)$ 2 revolutions of unit circle $\therefore +2\pi$

$$\Rightarrow (x + \frac{\pi}{3}) = \frac{\pi}{2}$$

$$\Rightarrow x = \frac{\pi}{2} - \frac{\pi}{3}$$

$$x = \frac{\pi}{6}$$

and $x = \frac{\pi}{6} + 2\pi = \frac{13\pi}{6}$

So Solutions are: $x = \frac{\pi}{6}$ and $\frac{13\pi}{6}$

6) $\cos(x - \frac{3\pi}{4}) = 0$ $x \in (0, 2\pi)$

$$\Rightarrow x - \frac{3\pi}{4} = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\Rightarrow x = -\frac{\pi}{2} + \frac{3\pi}{4}, \frac{\pi}{2} + \frac{3\pi}{4}, \frac{3\pi}{2} + \frac{3\pi}{4}$$

$$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

← discard as $> 2\pi$

So Solutions are: $x = \frac{\pi}{4}, \frac{5\pi}{4}$

7) $2\cos^2 x + 3\cos x + 1 = 0$ $x \in [0, 2\pi]$

Let $a = \cos x \Rightarrow 2a^2 + 3a + 1 = 0$

$$(2a+1)(a+1) = 0$$

$$a = -\frac{1}{2}, a = -1$$

So $\cos x = -\frac{1}{2}$

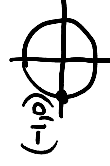
$$\Rightarrow x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$\Rightarrow x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

basic angle $\theta = \frac{\pi}{3}$
cos neg in Q2 $(\pi - \theta)$
+ Q3 $(\pi + \theta)$

$$\cos x = -1$$

$$x = \pi$$



x-coord -1 so
 $\theta = \pi$ at $(-1, 0)$

So the solutions are: $x = \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$

As $\sin \theta = 1$ the y-coord is 1

$\therefore$ or $\theta = \pi/2$ where point is $(0, 1)$

But we want bigger angles too $\therefore +2\pi$

(note if we added another 2π the answer would be $\frac{25\pi}{6}$ which is too big)



As $\cos \theta = 0$ the x-coord is 0

$\therefore$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ where point is either $(0, 1)$ or $(0, -1)$
but we could also look at $-\pi/2$.

If we add or subtract 2π then the answers would be outside domain